

Leveraging Transformer Models for Extractive Question Answering in Criminal Report Analysis

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Abstract. Transformer-based models have greatly advanced extractive question answering (EQA) in NLP, but their application to legal document analysis remains underexplored. This paper introduces a domain-adapted BERT model for EQA on Spanish-language robbery case reports from Ecuador’s Prosecutor’s Office (*Fiscalía General del Estado*). We used generative artificial intelligence to synthesize a SQuAD-format dataset tailored to robbery cases. The pre-trained BERT model was first evaluated in a zero-shot setting on this dataset, then further fine-tuned for the specific domain. The fine-tuned model outperformed the base version, achieving a final F1-score of 82.88. Our results demonstrate the effectiveness of transformer-based models, both in zero-shot and fine-tuned settings, for EQA tasks in legal language processing. These advancements highlight the potential to enhance information retrieval and optimize workloads in judicial systems, ultimately improving service delivery to communities.

Keywords: Legal Natural Language Processing · NLP · Natural Language Processing · transformers · LNLP · law.

1 Introduction

Transformer-based models have emerged as powerful tools in Natural Language Processing (NLP), enabling more accurate and efficient extraction of relevant information from complex texts such as criminal reports. These models utilize self-attention mechanisms to weigh the importance of different words in a sentence, allowing for better context understanding and improved performance in tasks like Question Answering (QA). By fine-tuning these models on domain-specific datasets, researchers can enhance their ability to discern pertinent details and provide precise answers to inquiries related to criminal incidents.

The implementation of Extractive Question Answering (EQA) systems in legal contexts not only enhances the efficiency of information retrieval but also addresses challenges related to data variability and complexity inherent in criminal reports. As these models are fine-tuned on diverse datasets, they can improve their generalizability across different types of legal documents, ensuring

that critical details are accurately captured regardless of nuances in language or structure [9]. Moreover, recent advancements in model architectures, such as the integration of dynamic masking techniques, have shown promise in further refining performance by enabling models to better adapt to varying levels of difficulty in questions posed about complex narratives [14]. This evolution highlights the potential for transformer-based models to facilitate more informed decision-making processes within the justice system, ultimately contributing to enhanced outcomes for legal practitioners and stakeholders alike.

The Prosecutor’s Office of Ecuador is responsible for handling a wide range of cases, including criminal investigations and public safety matters. Among these, robbery poses a significant challenge, with 67,705 reported incidents in 2024¹ (equivalent to 376 robberies per 100,000 inhabitants). The comprehensive analysis of these crime reports serves multiple critical functions: (1) it enables more effective investigations by revealing criminal patterns, (2) supports victims’ insurance claims for stolen property, and (3) provides essential data for public safety planning. Artificial Intelligence (AI) is particularly valuable in this context, as automated systems can enhance the efficiency of processing case information, alleviate backlogs, and ultimately improve service delivery to citizens.

Our previous research investigated robbery classification using state-of-the-art neural architectures, with a focus on fine-tuned language models. Extending this work, the current study investigates the application of EQA systems for processing criminal reports. The proposed methodology demonstrates significant potential to assist legal authorities by: (1) extracting actionable insights for criminal investigations, and (2) enabling comprehensive crime statistics analysis. To our knowledge, this represents the first application of transformer-based models utilizing a specially curated Spanish-language Stanford Question Answering Dataset (SQuAD) dataset derived from Ecuadorian legal documents.

This paper is organized as follows: Section 2 surveys existing applications of AI in legal domains, with a focus on question answering (QA) tasks and discusses important concepts such as Generative Artificial Intelligence (GAI) and Zero-Shot Learning (ZSL). Section 3 describes our methodology, including dataset construction and model training. Section 4 presents the comparative results of the ZSL model and the fine-tuned model. Finally, Section 5 summarizes our main findings and outlines opportunities for future work.

2 Literature Review

Large Language Models (LLMs) represent a category of Deep Learning (DL) models that integrate transformer architectures with large-scale pre-training on extensive text corpora. Trained in a self-supervised fashion, these models capture detailed linguistic patterns and can subsequently be fine-tuned to address domain-specific challenges in Legal NLP (LNLP). For instance, one influential

¹ Statistics reported by FGE with data cut-off date June 13, 2025

model is described in [4], and another is presented in [16], where OpenAI reported that the model achieved performance at the 90th percentile on the Uniform Bar Exam, a standardized examination used to evaluate the competency of prospective attorneys. However, subsequent analysis by [12] indicates that GPT-4’s actual performance ranges from the 48th to the 62nd percentile. These authors critique the imprecision and lack of transparency in OpenAI’s report, arguing that such shortcomings may hinder the development of safe AI and lead to misconceptions regarding the model’s true capabilities in handling complex legal tasks.

Legal Question Answering (LQA) is a complex task that entails addressing queries related to legal issues, necessitating not only the analysis of extensive legal corpora but also their nuanced interpretation [2]. A wide range of models and algorithms have been explored for QA in NLP, such as *Recurrent Neural Networks (RNNs)*, *Long Short-Term Memory (LSTM)*, and *Convolutional Neural Networks (CNN)*. Additionally, information retrieval approaches, including *Term Frequency-Inverse Document Frequency (TF-IDF)*, are frequently utilized. Both extractive and generative methodologies have demonstrated effectiveness in addressing NLP-based QA tasks [11].

It is important to clarify the primary distinction between these two approaches. EQA identifies and retrieves a span of text directly from the given context: Given a tuple composed of the question q and the context c , the model predicts the start and end positions of the answer. In contrast, Generative Question Answering (GQA) synthesizes responses using encoder-decoder architectures, generating answers in an autoregressive manner [11]. According to a systematic evaluation of QA models by [11], generative readers generally outperform extractive readers with longer contexts, providing more fluent and comprehensive answers, whereas extractive readers are more effective when the context is limited and exhibit better out-of-domain generalization.

2.1 Large Language Models and Generative AI

GAI refers to a class of AI models that learn the underlying data distribution $p_{data}(\mathbf{x})$ to generate novel samples (e.g., text, images, or audio) through an estimated model distribution $p_{model}(\mathbf{x})$ [5]. Modern AI models are considered Foundational Models, which represent a class of general-purpose models capable of handling diverse data modalities and supporting multiple AI tasks [7]. Notable examples include the Llama 3 family of models, which are foundational language models featuring multilingual capabilities, programming proficiency, reasoning skills, and tool integration [7]. Equation (1) represents the goal of GAI, which is to learn a model distribution $p_{model}(\mathbf{x})$ that approximates the true data distribution $p_{data}(\mathbf{x})$, where $\mathbf{x}$ represents observations from the data domain.

$$p_{model}(\mathbf{x}) \approx p_{data}(\mathbf{x}) \quad (1)$$

LLMs are a specialized subset of foundational AI and GAI models, with a primary focus on text-based tasks. These models leverage transformer architectures [7,13] and undergo extensive pre-training on textual corpora. Notably,

LLMs exhibit exceptional capabilities in natural language understanding and complex problem-solving (i.e., General Purpose Task Solver) [22], rendering them particularly effective for sophisticated linguistic tasks. The basic working of an LLMs can be thought of as a next-word prediction problem, where given a sequence of tokens $\mathbf{x}_{1:t} = (x_1, \dots, x_t)$, an LLM defines a conditional probability distribution over the vocabulary $\mathcal{V}$:

$$P(x_{t+1}|\mathbf{x}_{1:t};\theta) = \text{softmax}(\mathbf{W} \cdot \mathbf{h}_t + \mathbf{b}) \quad (2)$$

θ denotes the model parameters, and $\mathbf{h}_t$ is obtained by several layers of transformer blocks that process an embedded representation of the tokenized text:

$$\mathbf{h}_t = \text{Transformer}(\mathbf{x}_{1:t}; \theta) \quad (3)$$

2.2 Zero-Shot Learning

Traditional supervised learning in DL requires large labeled datasets, which can be difficult and expensive to obtain [19]. For instance, the SQuAD 2.0 benchmark dataset for EQA contains approximately 150,000 samples (100,000 answerable and 50,000 unanswerable) [17]. In this context, ZSL establishes a new paradigm in machine learning, where models trained on a subset of classes are expected to recognize new classes not seen during training [18]. Consequently, ZSL addresses the challenge posed by unlabeled data [19]. Moreover, pre-trained language models (e.g., BERT) that are trained on large-scale corpora achieve strong performance in ZSL by encoding linguistic patterns.

Transfer learning aims to utilize knowledge from a source domain to improve performance in a target domain with limited labeled data, typically through fine-tuning pre-trained language models [18]. However, this fine-tuning process can lead to catastrophic forgetting, where the model loses its original capabilities while adapting to new tasks. In contrast, ZSL does not require explicit labeling [8], thereby preserving the model’s original knowledge while enabling generalization to unseen classes.

In this research, we explore ZSL on the Legal EQA domain using a new set of samples from our dataset. Traditional applications in NLP include text classification, sentiment analysis, and Named Entity Recognition (NER) [19]. Our approach relies on the use of pre-trained language models such as BERT, which are powerful tools for ZSL. However, other ZSL techniques are: semantic embedding, attribute-based approaches, and generative [18,19]

2.3 Related Works

There is significant interest within both the scientific community and legal institutions in studying the applications of AI in this expansive field. The current state of the art in NLP with DL primarily focuses on the use of transformer-based architectures, which have demonstrated superior performance across various NLP tasks. However, improvements over this architecture are discussed

in [20], where a Bidirectional Long Short-Term Memory (BiLSTM) layer is integrated with the transformer to capture both global and sequential features. Although, their work is limited to natural language domain, the authors show that hybrid architectures with varied pooling strategies may improve state-of-the-art performance in QA.

In [21], the deployment of a RoBERTa Base QA classification model specifically for legal questions is discussed. It compares the performance of this transformer model against a traditional linear Support Vector Machine (SVM) in the legal domain, using the PRIVACYQA dataset. The results indicate that RoBERTa significantly outperforms the SVM, achieving a 31% improvement in F1-score and a 41% improvement in Mean Reciprocal Rank, demonstrating the effectiveness of transformers in enhancing answer retrieval for legal inquiries.

For information retrieval, [10] discuss the use of transformer-based approaches for legal information retrieval and entailment, specifically in the context of the COLIEE competition. For case law retrieval, a sentence-transformer model was employed to create numeric representations of case paragraphs, facilitating similarity assessments. As for the entailment task, a fine-tuned DeBERTa large language model was utilized, achieving third place among eight teams. These methods demonstrate the effectiveness of transformers in addressing LQA challenges.

Another area of interest is predicting legal judgments. In [6] six transformer-based models (BERT, XLNet, RoBERTa, DeBERTa, ELECTRA, and BigBird) are evaluated in predicting legal judgments using the ILDC dataset. It highlights the potential of these models in automating legal decision-making, with newer models achieving up to 80% accuracy compared to 76% for older models. This comparative analysis underscores the advancements in NLP technology and its applicability in the legal domain, suggesting a promising future for QA in law through transformer models.

Finally, in [15] the use of transformers to generate legal text is addressed, indicating their potential to assist in legal practice. The findings suggest that these models can effectively generate judicial opinions and help differentiate between human and machine-generated legal texts, which could be beneficial for tasks like drafting contracts and briefs, although direct application to QA is not discussed.

The reviewed studies indicate that transformer-based architectures have become the dominant approach in LNL, either through fine-tuning or architectural modifications. Applications include information retrieval, judgment prediction, and legal text generation. These trends highlight the relevance of the present work in addressing the LQA.

3 Methodology

In this paper, we present a BERT-based model fine-tuned for EQA on Spanish-language criminal reports. We started from a Spanish BERT architecture already pre-trained and optimized for QA tasks, then fine-tuned it to extract information from robbery crime narratives. Our methodology consists of three phases: (1) dataset generation, (2) ZSL pre-trained model performance, and (3) model fine-tuning.

3.1 Phase 1: Robbery Crime QA Dataset Generation

Data Collection and Preprocessing We constructed our dataset using a set of unlabeled robbery crime reports. The text preprocessing pipeline involved removing non-alphanumeric characters and converting all text to lowercase. Statistical analysis of the collected reports revealed an average length (mean) of 112 words, and an upper bound of 150 words. Consequently, we included reports x_i satisfying $82 \leq \Gamma(x_i) \leq 150$, where $\Gamma : x_i \rightarrow \mathbb{N}$ denotes the word count function for report x_i . This filtering criterion effectively removed statistical outliers while maintaining the core distribution. Figure 1 illustrates the word count distribution across the original dataset of 174,594 samples, presented through both histogram and box plot visualizations. Finally, we transformed this dataset into a QA format compatible with SQuAD using GAI.

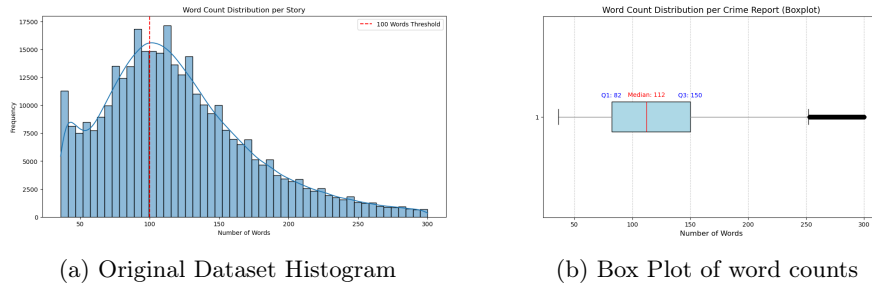


Fig. 1: Distribution of word counts in collected data presented as (a) histogram and (b) boxplot. The histogram shows a right-skewed distribution of crime reports. The interquartile range spans from 82 to 150 words, with a median of 112 words.

Generative AI Prompt Engineering For this research, we used Meta-LLAMA-3-8B-Instruct [1], an instruction-tuned variant of Meta’s LLAMA 3 8B parameter language model, optimized for conversational and task-oriented interactions with multilingual capabilities. Built on a decoder-only TRANSFORMER architecture, it leverages supervised fine-tuning (SFT) and reinforcement learning from human

feedback (RLHF) to align responses with user intent. The model is pre-trained on a diverse corpus of publicly available text data, with enhancements in reasoning, coding, and safety mitigation compared to its predecessors.

Prompt engineering is the process of crafting a suitable prompt to drive the LLMs to solve a complex task, using natural language text [22]. In this work, the LLM functions as an expert data science assistant with the goal of constructing a dataset in the SQuAD format in Spanish. For this purpose, the prompt receives the content of the crime report as the `context` and each of the questions from Table 1 as the `question`. The LLM was also instructed to retrieve a `json` format output with the variables: `context`, `question`, `answer_text`, `answer_start`, `answer_end`, and `impossible_find_answer`. The latter was calculated by finding the `answer_text` in the context. In the case that no answer was possible to be retrieved, either because there is no answer or because the LLM failed, the variable `impossible_find_answer` was set to `True`.

Table 1: Proposed Questions for the Dataset

Question
What objects were stolen?
On what date did the incident occur?
At what time did the robbery happen?
At what address or between which streets did the robbery/incident occur?
What was the dollar value of the stolen or taken objects?

Hugging Face’s Inference API processed samples at a speed of 6.4 samples per minute. A total of 1,544 samples were obtained after four hours. Our evaluation revealed two key limitations of the LLM: (1) incomplete retrieval of answers for certain queries, and (2) sporadic generation of irrelevant content. These results suggest that manual verification of the generated dataset can aid in improving quality. However, the engineered prompt allowed for the identification of problems in generated samples.

Dataset Quality After collecting the initial 1,544 samples, which generated 7,705 question-answer pairs at five questions per context, we conducted quality assessment using the `impossible_find_answer` flag. This process identified approximately 3,000 invalid cases, leading us to retain only samples where `impossible_find_answer` is `False`. Through histogram and box plot analysis of answer lengths, we established an upper limit of 17 words for valid responses. The final curated dataset consisted of 4,572 samples, all containing at least one word and meeting our validity criteria, suitable for model training and evaluation.

Dataset Train and Test Split The dataset was partitioned into training and testing sets, with 20% of the samples allocated for evaluation. The remaining

80% were used for training. This approach is a fundamental machine learning technique that helps to evaluate the model’s performance and generalization capability. Table 2 summarizes the sample sizes for each subset. The split was performed randomly.

Table 2: Dataset Train and Test Split

Partition	# Samples
Train	3,657
Test	915
Total	4,572

3.2 Phase 2: Zero Shot Learning Pre-Trained Model Performance

We selected the `mrm8488/bert-base-spanish-wwm-cased-finetuned-spa-squad2-es` model as the baseline pre-trained model due to its specialization for Spanish language processing in QA tasks. This model represents a distilled version of BETO (Spanish BERT) [3] that was further fine-tuned on the SQuAD-es-v2.0 dataset. The architecture follows the standard BERT-base [4] configuration, featuring:

- 12 Transformer blocks
- 768-dimensional hidden states
- 12 self-attention heads
- 384 tokens of sequence length

BETO was pretrained on an extensive Spanish corpus using Whole Word Masking. In preliminary testing without additional fine-tuning, the model achieved an F1-score of 81.70, demonstrating strong baseline performance for Spanish QA tasks.

3.3 Phase 3: Model Fine-Tuning

To fine-tune the pre-trained model, both the question and the answer need to be tokenized. As a result of this process, the indices indicating the start and end of the answer must be recalculated based on the tokenized embedding. In the case of long text, truncation should be applied only to the context, and a window of 128 tokens was used to avoid losing the answer during truncation. However, in this scenario, since most of the current samples have no more than 100 words in the context, this issue is unlikely to occur; nonetheless, it has been addressed in the program.

In order to train the model, the parameters shown in Table 3 were used. A small learning rate is recommended for fine-tuning in order to reuse the pre-training from the original model and adapt the new model to the target dataset.

Our results prove that it is possible to achieve a better performance on the target domain with the fine-tuned model rather than using the pre-trained model. Although the difference in performance is small, it could be related to the fact that we were not able to use the whole dataset.

Table 3: Training Parameters

Parameter	Value
Strategy	epoch
Learning rate	2×10^{-5}
Batch Size	64
Epochs	10
Weight Decay	0.01
FP16	True
Metric for best model F1 (eval)	

4 Results

In this section, we present the results of fine-tuning the model, along with the performance of the pre-trained model on the test set. This comparison was conducted to determine whether the fine-tuned model demonstrates any improvement over ZSL. Table 4 shows the performance of each learning approach used in this research.

Fine-tuning improves upon ZSL in terms of F1 score and exact match, with values of 82.88 and 81.70, respectively, indicating higher accuracy. The exact match metric measures the percentage of predictions that match any ground truth answer exactly. However, the pre-trained model used in ZSL shows lower evaluation loss and higher steps per second, suggesting less error and greater throughput. Thus, while fine-tuning yields better accuracy, the ZSL model is slightly less accurate but faster.

Our results suggest that increasing the number of samples for the fine-tuning approach may improve model performance. It is also important to note that the use of GAI enabled the synthesis of a training dataset through prompt engineering. According to [18] and [19], this represents another application of ZSL.

Table 4: Comparison of Pre-trained and Fine-tuned Model Performance

Metric	Zero Shot	Fine-tuning
eval_loss	0.9102	1.0088
eval_model_preparation_time	0.0077	0.0077
eval_exact	53.5519	55.7377
eval_f1	81.7041	82.8805
eval_total	915.0000	915.0000
eval_HasAns_exact	53.5519	55.7377
eval_HasAns_f1	81.7041	82.8805
eval_HasAns_total	915.0000	915.0000
eval_best_exact	53.5519	55.7377
eval_best_exact_thresh	0.0000	0.0000
eval_best_f1	81.7041	82.8805
eval_best_f1_thresh	0.0000	0.0000
eval_runtime	12.3982	12.8109
eval_samples_per_second	73.8010	71.4240
eval_steps_per_second	1.2100	1.1710
epoch	–	10.0000

5 Conclusions and Future Works

In this study, we demonstrated the effectiveness of transformer-based models for EQA within the context of criminal report analysis. Our fine-tuned model showed achieved an F1 score of 82.88, improving upon the pre-trained model’s zero-shot score of 81.70. This indicates the potential for enhanced performance through domain-specific fine-tuning. It also shows that ZSL can achieve a reasonable performance with a higher throughput.

The proposed methodology could be extended to other legal contexts in other countries, provided a pre-trained QA model in the target language is available. According to our results, a zero-shot approach achieves good performance. However, based on our experience, fine-tuning the model increases performance. Additionally, using GAI to create a SQuAD dataset can be adapted to the new target.

The application of our model to robbery crime reports could facilitate the extraction of crucial information, supporting legal practitioners in their investigations and reporting. However, the study also highlighted some limitations, including the model’s occasional inability to retrieve accurate answers and the need for manual verification of results.

Future work will address current limitations by exploring advanced techniques and incorporating manual dataset labeling by expert peers. Additional computational resources will be necessary for larger datasets. Integrating AI into the legal domain offers significant potential for improved case management and resource allocation, promoting a safer community. Further research should assess the impact of AI in the legal field, particularly regarding data and methodological biases. Exploring other NLP approaches, such as Legal Legal NER and Legal

Document Summarization (LDS), within the Ecuadorian context is also recommended. Continued research and collaboration are encouraged to refine these methods and maximize AI's benefits in supporting the justice system.

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